

iPad tops tablet traffic online

According to comScore, iPad and iPad2 together account for over 89 percent of tablet internet use globally and 97 percent in the US

Archos enters Honeycomb bandwagon

Archos has unveiled its 80 G9 and 101 G9 Android 3.1 tablets housing a 1.5GHz ARM Cortex A9 processor, support for Flash and 3G ready

Hard Drives Test

SOLID STATE DRIVES

Solid State Drives (SSDs) have been around for a while now and even though they cannot match upto traditional hard drives in terms of cost per GB, their importance cannot be undermined. Thankfully capacities are increasing and prices are dropping.

But there are certain things that need consideration before purchasing an SSD such as TRIM support, better controllers and use case scenarios. SSDs are known to have great read speeds, but the same cannot be said about write speeds. Sequential write speed is also affected by the capacity of the drive because higher density drives have large number of NAND flash chips which allows the controller to access more storage simultaneously.

Another point worth noting is that even though flash memory lasts longer than traditional magnetic recording, an SSD can only handle a finite number of write operations, beyond which the integrity of the data is questionable.

SSD controllers are very important in an SSD as they basically manage the flow of data between the NAND flash chips and the computer, controlling of writing methods, etc. Speaking of which the Transcend TS32GSSD25S-M houses the older JMicron JMF602B



Kingston SSDNow V+100 won the Best Performer for its fabulous Read and Write speeds

controller which is known for issues like stuttering. Although the box rates speeds of upto 260 MB/s (read we are assuming as nothing is mentioned) our testing showed read speeds around 150 MB/s. Random write speed was quite appalling at 17 MB/s. ADATA S596 performed marginally better thanks to the upgraded JMicron JMF612 controller, giving much higher read speeds, but the write speeds were still around 60MB/s for sequential write and around 20

MB/s for random write. While the Transcend 32GB drive is not recommended due to the older controller and absence of TRIM support, the ADATA S596 drive can be considered by those looking at faster application loading.

Apacer AS206 housing the SandForce SF1200 controller gave better write performance than the ADATA and Transcend SSDs.

Kingston SSDNow V100 and V+100 were the only SSDs that left us impressed. Although the V100 is an older SSD, the newer V+100 gave improved

write speeds for both sequential and random writes (over 40 percent improvement).

Verdict

There was no competition among the SSDs really, with the Kingstons scoring in performance. Kingston SSDNow V+100 was judged as the Best Performer as well as the Best Buy. It got a Best Buy despite a high price of ₹119 per GB as all the other competing drives had a negative price to performance ratio. If we had to still recommend an SSD apart from Kingston, we would stick with Apacer. **d**

RAID 0 PERFORMANCE IMPROVEMENT

We had set out to include RAID 0 readings as well in our testing, but despite a detailed mail explaining the testing procedure, only Seagate sent us two units of most of their drives. We still went ahead with performing RAID 0 on two drives, the WD Velociraptor WD3000HLFS (300GB) and Seagate Momentus

Before we analyse the scores, a quick primer on RAID 0. RAID is the acronym for Redundant Array of Inexpensive Disks, which combines two (or more) hard drives to form one single drive. This may result in superior performance or greater reliability or both. We decided to do RAID 0 as we wanted to check the speed boost that one can derive with this configuration. RAID 0 uses a method called Striping, wherein a single chunk of data is split into two and sent equally to both the drives. For instance in the data stream 100110, this is how the data will be split among drive A and drive B: ABABAB such that drive A has 101 and drive B has 010. As a result of this performance improves, as what would otherwise have taken 6 steps for a single drive is done in only 3 steps for two drives working in parallel. But there is a cost behind this gain in performance: if one drive fails, your data is lost, as the other drive only has the other half of the data, which doesn't have any meaning. The advantages are higher performance and increased capacity.

RAID 1 configuration makes use of Mirroring. Coming back to our data stream (100110), in RAID 1, both drive A and drive B will have the complete string 100110 ie. In place of 6 steps for a single drive it will take 12 steps for two drives. This means that if one of the drive fails, the other drive will still have all the data.

There are more RAID configurations which require more than two drives which we will discuss some other time.

Coming to our performance scores, here are the charts. The idea behind this exercise was to explain, how you can achieve great speeds by having two inexpensive drives in RAID 0. Of course, do not forget to get the crucial data backed up.

